

Assignment – 7

Name: Shashank Raj

Roll: 25SCS1003001227

Sec: 2CSE14

Q) Bank Account Management System :

Create a class named BankAccount with the following private data members:

accountNumber

accountHolderName

balance

```
class BankAccount {  
  
    // Private data members  
  
    private String accountNumber;  
  
    private String accountHolderName;  
  
    private double balance;  
  
    // Parameterized constructor  
  
    public BankAccount(String accountNumber, String accountHolderName, double balance) {  
  
        setAccountNumber(accountNumber);  
  
        setAccountHolderName(accountHolderName);  
  
        setBalance(balance);  
  
    }  
  
    // Getter and Setter for Account Number  
  
    public String getAccountNumber() {  
  
        return accountNumber;  
  
    }  
  
    public void setAccountNumber(String accountNumber) {  
  
        if (accountNumber != null && !accountNumber.isEmpty())  
  
            && accountNumber.length() >= 5) {  
  
                this.accountNumber = accountNumber;  
  
            } else {  
  
                System.out.println("Invalid Account Number");  
  
            }  
  
    }  
  
    // Getter and Setter for Account Holder Name
```

```
public String getAccountHolderName() {  
    return accountHolderName;  
}  
  
public void setAccountHolderName(String accountHolderName) {  
    if (accountHolderName != null && !accountHolderName.isEmpty()  
        && accountHolderName.length() >= 3) {  
        this.accountHolderName = accountHolderName;  
    } else {  
        System.out.println("Invalid Account Holder Name");  
    }  
}  
  
// Getter and Setter for Balance  
public double getBalance() {  
    return balance;  
}  
  
public void setBalance(double balance) {  
    if (balance >= 0) {  
        this.balance = balance;  
    } else {  
        System.out.println("Invalid Balance");  
    }  
}  
  
// Deposit method  
public void deposit(double amount) {  
    if (amount > 0) {  
        balance += amount;  
        System.out.println("Deposited Amount: " + amount);  
    } else {  
        System.out.println("Invalid Deposit Amount");  
    }  
}  
  
// Withdraw method  
public void withdraw(double amount) {  
    if (amount > 0 && amount <= balance) {  
        balance -= amount;  
        System.out.println("Withdrawn Amount: " + amount);  
    }  
}
```

```
    } else {

        System.out.println("Insufficient Balance");

    }

}

// Display account details

public void displayAccountDetails() {

    System.out.println("Account Number: " + accountNumber);

    System.out.println("Account Holder Name: " + accountHolderName);

    System.out.println("Available Balance: " + balance);

}

}

// Main class

public class Question {

    public static void main(String[] args) {

        BankAccount account = new BankAccount(

            "ACC1003001",

            "Shashank Raj",

            50000.0

        );

        account.displayAccountDetails();

        System.out.println("\nDepositing: 2000");

        account.deposit(2000);

        System.out.println("Withdrawal: 5000");

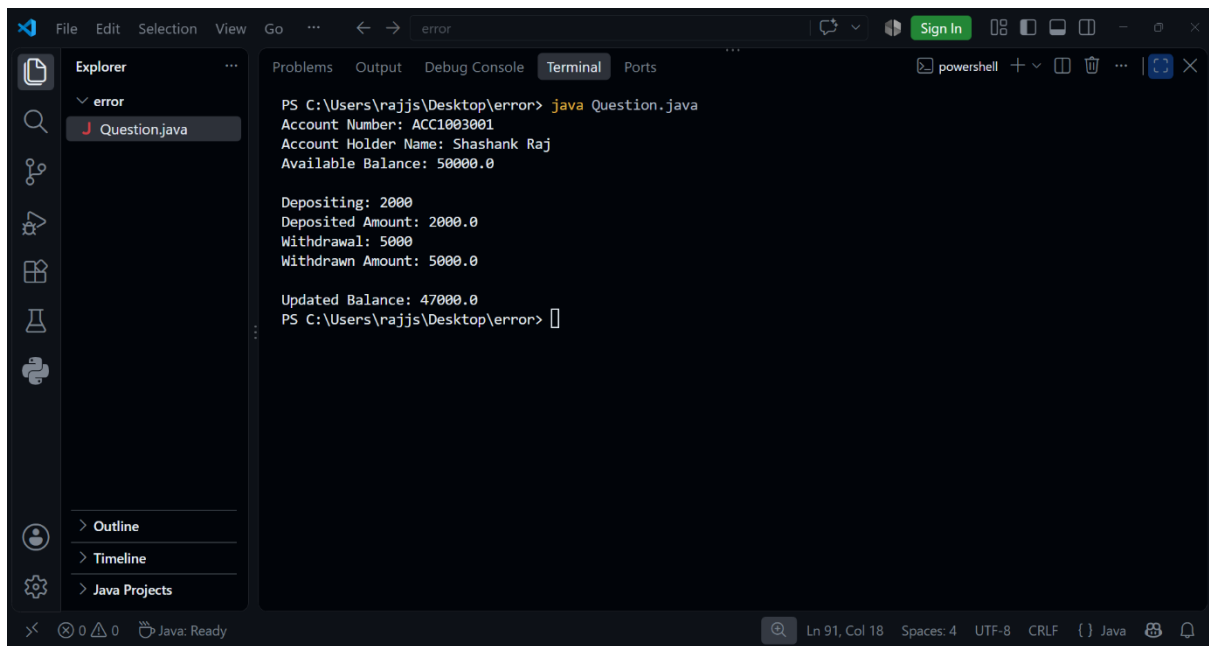
        account.withdraw(5000);

        System.out.println("\nUpdated Balance: " + account.getBalance());

    }

}
```

OUTPUT-



The screenshot shows a Java IDE with a dark theme. The Explorer panel on the left shows a project named 'error' containing a file 'Question.java'. The Terminal panel on the right displays the output of running 'java Question.java'. The output shows account details and a balance update.

```
PS C:\Users\rajjs\Desktop\error> java Question.java
Account Number: ACC1003001
Account Holder Name: Shashank Raj
Available Balance: 50000.0

Depositing: 2000
Deposited Amount: 2000.0
Withdrawal: 5000
Withdrawn Amount: 5000.0

Updated Balance: 47000.0
PS C:\Users\rajjs\Desktop\error> 
```

The status bar at the bottom indicates 'Ln 91, Col 18', 'Spaces: 4', 'UTF-8', 'CRLF', and the language is set to 'Java'.